Maximal-Tensor Multiplication Does Not Extend on the CAR Algebra

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22 July 2026

Status. Candidate counterexample, likely valid, for human review. The proof is complete; novelty is supported only by the bounded search in Section 6.

1 Source question

Let A be a unital C^*-algebra. Lesch writes

$$\mu_n : A^{\odot(n+1)} \longrightarrow A, \qquad \mu_n(a_0 \otimes \cdots \otimes a_n) = a_0 a_1 \cdots a_n,$$

and asks whether it extends continuously to the maximal C^*-tensor product $A^{\otimes_{\max}(n+1)}$ [2, Section 3.1]. He says that he suspects interesting examples where it does not extend, but that an expert poll was inconclusive. Liu later describes the extension question as "still inconclusive" and cites Lesch [1, Section 2, PDF p. 5].

$$p_\pi^\gamma : A^{\otimes_\gamma n+1} \to A^{\otimes_\pi n+1}. \tag{2.11}$$

Accordingly, $A^{\otimes_\pi n+1}$ is a quotient of $A^{\otimes_\gamma n+1}$. The two topological tensor products come with somehow mutually exclusive features that are essential for constructing functional calculi. The maximal C^*-tensor product $\otimes_\pi$ has the advantage that the completion $A^{\otimes_\pi n}$ is still a C^*-algebra. For the C^*-algebra $C(X_h)$ in Eq. (2.8), the maximal $(n+1)$-fold tensor product is isomorphic to the algebra of continuous functions on the $(n+1)$-fold Cartesian product of X_h: $C(X_h)^{\otimes_\pi n+1} \cong C(X_h^{n+1})$. Similar to Eq. (2.8), for $h^{(0)}, \ldots, h^{(n)} \in A^{\otimes n+1}$[2], we have the joint spectral measure (a $*$-isomorphism)

$$\Psi_\pi : C(X_h^{n+1}) \to C^*(1, h^{(0)}, \ldots, h^{(n)}) \subset A^{\otimes_\pi n+1}, \tag{2.12}$$

where the C^*-algebra generated by $1, h^{(0)}, \ldots, h^{(n)}$ is sitting inside the maximal C^*-tensor product. The drawback of the maximal C^*-tensor product is that whether the multiplication map $\mu_n : A^{\otimes n} \to A$ can be continuously extended to $A^{\otimes_\pi n} \to A$ is still inconclusive [3]. thus we don't know whether the map ι in Eq. (2.2) extends continuously to $A^{\otimes_\pi n+1} \to L_{\text{cont}}(A^{\otimes_\pi n}, A)$. To make use of such features, we turn to the projective one $A^{\otimes_\gamma n+1}$ that behaves well when extending the multiplication map μ_{n+1}, in particular, ι in Eq. (2.2) admits a continuous extension

$$\iota : A^{\otimes_\gamma n+1} \to L_{\text{cont}}(A^{\otimes_\pi n}, A). \tag{2.13}$$

[2] Strictly speaking, here $h^{(j)}$ means the elementary tensor $1 \otimes \cdots h \cdots \otimes 1$ before applying the ι map, cf. Eq. (2.3).

[3] The author's knowledge on this issue is limited to the discussion in [Les17, §3]

Liu's displayed multiplication map uses a shifted number of tensor factors. To avoid this notational issue, below m_k always means multiplication on exactly k algebraic factors. Thus Lesch's μ_n is m_{n+1}.

The same sources consider the contraction map

$$\iota : A^{\odot(n+1)} \longrightarrow L\big(A^{\odot n}, A\big),$$

where

$$\iota(a_0 \otimes \cdots \otimes a_n)(b_1 \otimes \cdots \otimes b_n) = a_0 b_1 a_1 \cdots b_n a_n.$$

They ask whether a suitable joint maximal-tensor extension exists.

2 Definitions

For C^*-algebras B and D, $B \odot D$ denotes their algebraic tensor product, while $B \otimes_{\min} D$ and $B \otimes_{\max} D$ denote the minimal and maximal C^*-completions. The minimal tensor product is injective: inclusions of C^*-algebras induce isometric inclusions of minimal tensor products. A C^*-algebra A is nuclear if the canonical maximal-to-minimal quotient is an isomorphism for $A \odot B$ and every C^*-algebra B; see, for example, [3, Chapters 2–3].

The CAR algebra is

$$\mathcal{A} = M_{2^\infty} = \varinjlim_r M_2^{\otimes r}.$$

It is unital, separable, simple, and nuclear. Its rth finite stage is a unital copy of M_d, where $d = 2^r$.

3 Proof intuition

Matrix multiplication hides a dimension factor that the spatial tensor norm does not see. In $M_d \odot M_d$, the tensor $z_d = \sum_j e_{1j} \otimes e_{j1}$ merely moves one d-dimensional coordinate subspace isometrically to another, so its tensor norm is one. Multiplication, however, collapses all d summands onto the same matrix unit and produces de_{11}. The CAR algebra contains such matrix blocks for unbounded d. Nuclearity is the bridge from this spatial calculation to the maximal tensor norm asked about in the source.

4 Counterexample

We first record the reusable obstruction.

Proposition 1 (Unbounded matrix-block criterion)**.** *Let A be a unital nuclear C^*-algebra. Suppose that A contains C^*-subalgebras isomorphic to M_d for an unbounded set of integers d. Then, for every $k \geq 2$, the algebraic multiplication map*

$$m_k : A^{\odot k} \longrightarrow A, \qquad m_k(a_1 \otimes \cdots \otimes a_k) = a_1 \cdots a_k,$$

is unbounded for the maximal C^*-tensor norm. Hence it has no continuous extension to $A^{\otimes_{\max} k}$.

Proof. Fix one of the subalgebras $M_d \subseteq A$ and choose standard matrix units $(e_{ij})_{i,j=1}^d$. In $M_d \odot M_d$ define

$$z_d = \sum_{j=1}^{d} e_{1j} \otimes e_{j1}.$$

Let $(\xi_j)_{j=1}^d$ be the standard basis of $\mathbb{C}^d$. In the spatial representation on $\mathbb{C}^d \otimes \mathbb{C}^d$,

$$z_d(\xi_p \otimes \xi_q) = \begin{cases} \xi_1 \otimes \xi_p, & q = 1, \\ 0, & q \neq 1. \end{cases}$$

Thus z_d is a partial isometry from $\mathbb{C}^d \otimes \mathbb{C}\xi_1$ onto $\mathbb{C}\xi_1 \otimes \mathbb{C}^d$, and

$$\|z_d\|_{M_d \otimes_{\min} M_d} = 1. \tag{1}$$

By injectivity of the minimal tensor product, the inclusion $M_d \subseteq A$ preserves the norm in (1). Since A is nuclear, maximal and minimal tensor powers of A coincide. Consequently,

$$\|z_d\|_{A \otimes_{\max} A} = 1.$$

But binary multiplication gives

$$m_2(z_d) = \sum_{j=1}^{d} e_{1j} e_{j1} = de_{11}, \qquad \|m_2(z_d)\| = d. \tag{2}$$

As the available dimensions are unbounded, m_2 is unbounded.

For $k > 2$, put

$$z_d^{(k)} = z_d \otimes 1_A^{\otimes(k-2)} \in A^{\odot k}.$$

Tensoring by identities preserves the C^*-norm, so $\|z_d^{(k)}\|_{A^{\otimes_{\max} k}} = 1$, whereas $m_k(z_d^{(k)}) = de_{11}$. Hence every m_k, $k \geq 2$, is unbounded. $\square$

Theorem 1 (CAR counterexample)**.** *For the CAR algebra $\mathcal{A} = M_{2^\infty}$ and every $n \geq 1$, Lesch's map*

$$\mu_n : \mathcal{A}^{\odot(n+1)} \longrightarrow \mathcal{A}$$

does not extend continuously to $\mathcal{A}^{\otimes_{\max}(n+1)}$. Consequently, the contraction map in Liu and Lesch has no continuous extension

$$\iota : \mathcal{A}^{\otimes_{\max}(n+1)} \longrightarrow L_{\text{cont}}(\mathcal{A}^{\otimes_{\max} n}, \mathcal{A})$$

that agrees with the algebraic contraction.

Proof. The CAR algebra is nuclear and contains the unital finite stages M_{2^r} for all r. Proposition 1, with $k = n + 1$, proves the first assertion.

For the second, suppose such an extension of ι existed. Evaluation at $1_A^{\otimes n}$ is a bounded map from $L_{\text{cont}}(\mathcal{A}^{\otimes_{\max} n}, \mathcal{A})$ to $\mathcal{A}$. On algebraic tensors its composition with ι is exactly

$$a_0 \otimes \cdots \otimes a_n \longmapsto a_0 a_1 \cdots a_n = \mu_n(a).$$

The composition would therefore give the forbidden bounded extension of μ_n. $\square$

This supplies a separable, unital, simple, nuclear example. It also shows that the failure is simultaneous in every nontrivial multilinear degree.

5 Verification and limitations

The script `code/verifier.py` constructs z_d as an operator on $\mathbb{C}^d \otimes \mathbb{C}^d$ and checks, for $d = 2, 4, 8, 16$, that

$$\|z_d\| = 1, \qquad z_d z_d^* z_d = z_d, \qquad m_2(z_d) = de_{11}.$$

It runs with:

```
conda run --no-capture-output -n sandbox python code/verifier.py
```

All residuals are zero and the computed multiplication norms are $2, 4, 8, 16$. This check guards against an index or tensor-order error; the proof for all $d = 2^r$ is the displayed matrix-unit calculation.

The theorem answers the extension question negatively in general. It does not classify all C^*-algebras for which multiplication is bounded, and it does not settle the case of the irrational rotation algebra occurring in Liu's geometric applications.

6 Bounded novelty check

The search on 22 July 2026 covered the run's registry, solution, attempt, and proof-gap indexes; the exact source ids and titles; and web/arXiv searches combining "multiplication map" and "maximal C^*-tensor product" with CAR, UHF, nuclear, minimal tensor product, and subhomogeneous.

Vaes and Van Daele state that multiplication generally need not extend to the minimal C^*-tensor product [4, Introduction]. Van Daele's later survey points to growth of the multiplication norms of M_d as $d \to \infty$ [5, Remark 3.3]. These are close precursors, but the inspected passages do not state the fixed CAR/UHF example or transfer it to the maximal norm through nuclearity. A 2021 MathOverflow discussion gives a spatial-tensor obstruction for $B(H)$, which is not by itself a maximal-norm answer. No explicit CAR/UHF maximal-tensor counterexample was found.

This supports, but does not certify, novelty. Because the construction is an elementary synthesis of standard facts, the priority claim requires a deeper operator-algebra literature review. The mathematical validity of the counterexample does not depend on that novelty assessment.

7 Human-review recommendation

Send the packet to an operator-algebra or C^*-tensor-product specialist. The proof audit should focus on the injective minimal-tensor inclusion and the nuclear maximal-to-minimal identification. The larger review burden is a specialist novelty search, particularly in the Hopf-C^*-algebra and operator-space literature.

References

[1] Y. Liu, *Hypergeometric function and Modular Curvature I. Hypergeometric functions in Heat Coefficients*, arXiv:1810.09939 (2018), `https://arxiv.org/abs/1810.09939`.

[2] M. Lesch, *Divided differences in noncommutative geometry: rearrangement lemma, functional calculus and expansional formula*, J. Noncommut. Geom. 11 (2017), 193–223; arXiv:1405.0863, `https://arxiv.org/abs/1405.0863`.

[3] N. P. Brown and N. Ozawa, *C^*-Algebras and Finite-Dimensional Approximations*, Graduate Studies in Mathematics 88, American Mathematical Society, 2008.

[4] S. Vaes and A. Van Daele, *Hopf C^*-algebras*, Proc. London Math. Soc. 82 (2001), 337–384; arXiv:math/9907030, `https://arxiv.org/abs/math/9907030`.

[5] A. Van Daele, *From Hopf algebras to topological quantum groups: a short history, various aspects and some problems*, Banach Center Publ. 120 (2020), 179–223, doi:10.4064/bc120-14.